Rats or Cats which sleeps longer?

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Introduction

In this analysis, we will address the question of whether rats (Rodentia) and cats (Carnivora) have similar number of sleep hours.

Data

The data is contained in the `msleep` dataset in the `tidyverse` package. This is loaded:

```
library(tidyverse)
data("msleep")
```

The data consists of 83 Species with 11 variables measured on it.

Filter data

We are only interested in the variables:

- **order:** which is a categorical nominal variable and gives the Order of the Species.
- **sleep_total:** which is a quantitative continuous variable and gives the total amount of sleep in hours.

As well, we are only interested in Species that are in the order Carnivora or Rodentia. The following code filters the dataset:

```
rats_cats  <- msleep %>%
  filter(order %in% c("Carnivora", "Rodentia")) %>%
  select(order, sleep_total)
```

Basic summary statistics

Table **??** gives the summary statistics for each group.

Table 1: Frequency table of Order

.	Freq
Carnivora	12
Rodentia	22

Side-by-side boxplots

Figure **??** gives the side-by-side boxplots of sleep total for each order.

Two-sample t-test

The sample mean for each group was used. To recap the sample mean for each group is

$$\bar{Y}_i = \sum_{i=1}^{n_i} Y_{ij},$$

where n_i is the number in the ith group, and Y_{ij} is the jth observation in the ith group.

The appropriate null and alternative hypotheses are

$$\begin{aligned} H_0 &: ? \\ H_a &: ? \end{aligned}$$

where μ_1 is the mean of Carnivora and μ_2 is the mean of Rodentia.

The test statistic is given by $$

$$

> INSERT test statistic formula here...

We perform a two-sample t-test with

Conclusion

> 95% confidence interval and P-value

Session info

```
# INSERT session info here
```